The Effective Number of Parameters

- The concept of number of parameters can be generalized to models where regularization is used in the fitting.
- A linear fitting method is one for which:

$$\hat{\boldsymbol{y}} = S\boldsymbol{y}, \qquad \text{where } S \in \mathbb{R}^{N\times N} \text{ depends only on the design matrix } X.$$

- For instance, for least squares, we have:

$$\hat{\boldsymbol{y}} = X\hat{\beta}_{LS} = X(X'X)^{-1}X'\boldsymbol{y} = P_X\boldsymbol{y},$$

  where P_X is the hat matrix.
- The effective number of parameters is defined as $df(S) = \mathrm{Tr}(S)$.
- For instance, for least squares with p inputs s.t. $X \in \mathbb{R}^{N\times p}$ is full rank, we know that $\mathrm{Tr}(P_X) = p$.
- For a model of the sort $Y = f(\boldsymbol{x}) + \varepsilon$ with $\mathrm{Var}(\varepsilon) = \sigma^2$, we already showed that

$$\sum_{i=1}^{N} \mathrm{Cov}(\hat{y}_i, y_i) = \mathrm{Tr}(S)\,\sigma^2$$

  (we did so for a linear fit with $Y = X\beta + \varepsilon$, but the idea is the same.)
- This motivates the more general definition:

$$df(\hat{\boldsymbol{y}}) = \mathrm{Tr}(S) = \frac{\sum_{i=1}^{N} \mathrm{Cov}(\hat{y}_i, y_i)}{\sigma^2}$$